

Ontology Engineering

Session 4

Assumptions

- Febraury 18
 - **Protégé** installed, and basic knowledge of its functionalities
- February 25
 - **formal taxonomy** of road signs as an IS-A class hierarchy, written in OWL2
 - conformant to Part I, Annex 1, of Vienna Convention (document “Conv_road_signs_2006v_EN.pdf”)
- March 4
 - **extended taxonomy**, aligned with the category labels of the RDF knowledge graph of the Mapillary Traffic Sign Dataset (file “MTSD.ttl”)
- March 18
 - **object properties and data properties**
 - conformant to Part I, Annex 1, of Vienna Convention (document “Conv_road_signs_2006v_EN.pdf”)

A Simple Ontology Development Methodology

1. Determine the domain and scope of the ontology
2. Consider reusing existing ontologies
3. Enumerate terms in the ontology
4. Define the classes and the class hierarchy
5. Define the properties of classes
6. Define the characteristics of properties
7. Create instances

roadsign (http://www.iiia.csic.es/~marco/kr/roadsign.owl) : [/Users/schorlemmer/Documents/Codex/KR 2026/roadsign.owl]

roadsign (http://www.iiia.csic.es/~marco/kr/roadsign.owl)

hasBorderColour

Active ontology x Entities x Individuals by class x DL Query x

Annotation propertiesDatatypesIndividuals

ClassesObject propertiesData properties

Object property hierarchy: hasBorderColour

owl:topObjectProperty

foaf:depicts

hasBorderColour

hasGroundColour

hasShape

hasSymbol

hasSymbolColour

relevantToRoadUser

Asserted

AnnotationsUsage

Annotations: hasBorderColour

Annotations

Character

Description: hasBorderColour

☐ Functional

☐ Inverse functional

☐ Transitive

☐ Symmetric

☐ Asymmetric

☐ Reflexive

☐ Irreflexive

Equivalent To

SubProperty Of

Inverse Of

Domains (intersection)

Ranges (intersection)

Disjoint With

SuperProperty Of (Chain)

RoadSign

Colour

To use the reasoner click Reasoner > Start reasoner

☒ Show Inferences

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roadsign (http://www.iiia.csic.es/~marco/kr/roadsign.owl)

🔍

symbol_value

Active ontology × Entities × Individuals by class × DL Query ×

Annotation properties

Datatypes

Individuals

Classes

Object properties

Data properties

Data property hierarchy: symbol_value

owl:topDataProperty

border_colour

ground_colour

shape

symbol

symbol_colour

symbol_value

Asserted

Annotations

Usage

Annotations: symbol_value

Annotations

Characterist

Description: symbol_value

☐ Functional

Equivalent To

SubProperty Of

Domains (intersection)

Ranges

Disjoint With

RoadSign

xsd:integer

To use the reasoner click Reasoner > Start reasoner

☒ Show Inferences

Necessary and Sufficient Conditions

A “STOP” sign **necessarily** is octagonal:

- FOL: $\forall x . \text{StopSign}(x) \rightarrow \text{hasShape}(x, \text{OCTAGON})$
- DL: $\text{StopSign} \sqsubseteq \text{hasShape} : \text{OCTAGON}$
StopSign subClassOf (hasShape value OCTAGON) (Manchester Syntax)



A road sign, to be a stop sign, is **sufficient** to be octagonal, with red ground colour, white border colour, and “STOP” written on it.

- FOL:
 $\forall x . (\text{RoadSign}(x) \wedge \text{hasShape}(x, \text{OCTAGON}) \wedge \text{hasGroundColour}(x, \text{RED}) \rightarrow \text{StopSign}(x))$
- DL: $\text{RoadSign} \sqcap (\text{hasShape} : \text{OCTAGON}) \sqcap (\text{hasGroundColour} : \text{RED}) \sqsubseteq \text{StopSign}$

Manchester Syntax

Helper Manchester Syntax

This table shows the conversions between DL syntax and Manchester syntax for these constructors:

Constructor	DL syntax	Manchester syntax
Bottom	$\perp$	Nothing
Top	$\top$	Thing
Intersection	$C \sqcap D$	C AND D
Union	$C \sqcup D$	C OR D
Complement	$\neg C$	NOT C
Existential restriction	$\exists R.C$	R SOME C
Universal restriction	$\forall R.C$	R ONLY C
Min. cardinality restriction	$\geq n R.C$	R MIN nC
Max. cardinality restriction	$\leq n R.C$	R MAX nC
Eq. cardinality restriction	$= n R.C$	R EXACTLY nC
Self restriction	$\exists R.\text{Self}$	R SELF
Value restriction	$\exists R \{a\}$	R VALUE {a}
Property chain	$R \cdot P$	R O P
Inverse property	R^-	INVERSE R
Property complement	$\neg R$	NOT R

This table shows the conversions between DL syntax and Manchester syntax for these axioms:

Axiom	DL syntax	Manchester syntax
Subclass of	$C \sqsubseteq D$	C SubClassOf D
Equivalent Class	$C \equiv D$	C EquivalentTo D
SubPropertyOf	$R \sqsubseteq P$	C SubPropertyOf D
Equivalent property	$R \equiv P$	R EquivalentTo P

Ok

Definitions

DangerWarningSign $\equiv$ RoadSign

- (hasShape : TRIANGLE)
- (hasGroundColour : WHITE)
- (hasBorderColour : RED)



Assignment

- Add to your ontology:
 - **property characteristics** (for object and data properties)
 - **definitions** for *all* road-sign classes of the main hierarchy
- Do this using the appropriate prompts for LLM-based ontology development
- You may accompany the prompt with examples of characteristics and definitions, and also with competency questions

Remember to Keep a Record of...

- For each engineering step:
 - LLMs used
 - prompt (or series of prompts)
 - revised output file
- This information has to be provided in the final submission

Optional Submission

- List of **competency questions** that is guiding your ontology developments
- Current state of your **road-sign ontology** with the main taxonomy of road signs, MTDS categories, object and data properties (with their characteristics) and road-sign definitions